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Kubernetes Deployment Guide — EKS Production Clusters

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This guide documents NovaTech's EKS cluster configuration, namespace conventions, resource quotas, HPA/KEDA autoscaling policies, and the GitOps deployment workflow via ArgoCD. All Project Atlas services must be deployed using the novatech-helm-chart base chart with mandatory security contexts and network policies.

1. Cluster Layout

Production runs two EKS clusters: **us-east-1 (primary)** and **eu-west-1 (DR)**. Node groups are split by workload: compute-optimised (c6i.4xlarge) for CPU-bound services, memory-optimised (r6g.2xlarge) for Qdrant and Redis, and GPU (g4dn.xlarge) for the reranker and embedding inference services.

Namespace	Services	Resource Quota (CPU/Mem)
retriever	retriever-api, qdrant	32 / 128 Gi
embed	embedder-svc, celery-embed	64 / 64 Gi
gateway	api-gateway, auth-proxy	16 / 32 Gi
monitoring	prometheus, grafana, tempo	8 / 16 Gi
ingress	nginx-ingress, cert-manager	4 / 8 Gi

2. Autoscaling

CPU-bound services use HPA with target CPU utilisation of 65%. The Celery worker fleet uses KEDA with a Redis queue-length trigger: 1 worker per 50 pending tasks, min=2, max=20 per queue. Qdrant and stateful services are not autoscaled; capacity is provisioned via quarterly capacity planning reviews.

3. Security Contexts

All pods must run as non-root (runAsNonRoot: true, runAsUser: 1000). Read-only root filesystems are enforced for stateless services. Privileged containers are blocked by OPA Gatekeeper policy. Image signatures are verified via Cosign before deployment.

- Pod Security Standards: Restricted profile enforced cluster-wide.
- Network Policies: deny-all default; explicit allow rules per namespace.
- Secrets: all environment secrets sourced from AWS Secrets Manager via ESO.
- Image scanning: Trivy scans all images in CI; HIGH CVEs block merge.